

A Machine-Checked Obstruction in the Calibration Step of the Switching-by-Weakness Route to $P \neq NP$

The MeTTapedia formalization effort

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Abstract

Ben Goertzel’s manuscript $P \neq NP$: *A Non-Relativizing Proof via Quantale Weakness and Geometric Complexity* (v12) proposes to separate P from NP by an information-theoretic *quantale* argument. On an efficiently samplable ensemble $\mathcal{D}_m$ of masked random Unique-SAT instances with a Valiant–Vazirani isolation layer, a *Switching-by-Weakness* normal form (Theorem 4.2) reduces every short global decoder, after a short wrapper, to an $O(\log m)$ -local per-block rule on the post-switch input $\mathbf{u}_i = (\mathbf{z}, a_i, b)$; sign-invariant neutrality (Theorem 5.1) and template sparsification (Theorem 5.10) then pin each local rule near $1/2$, independence across $\Theta(m)$ blocks yields per-program success $2^{-\Omega(t)}$, hence a distributional lower bound $K^{\text{poly}} \geq \eta t$, which clashes with the constant-length self-reduction upper bound under $P = NP$ (Proposition 7.2). The load-bearing step is the *calibration/success-domination* link (Theorem 4.2, Corollary 4.6, Lemma 4.8; Appendix A.5, Lemma A.17), which derives the exchangeability of (X_i, Y_i) at a fixed local input $\mathbf{u}$ from the promise-preserving involution T_i . We record a machine-checked finding of the MeTTapedia formalization effort: the involution T_i acts on the post-switch input $\mathbf{u}_i = (\mathbf{z}, a_i, b)$ by $b \mapsto b \oplus a_i$, so it preserves only the *reduced projection* $(\mathbf{z}, a_i)$, and preserves the *full* input $(\mathbf{z}, a_i, b)$ if and only if $a_i = 0$. Consequently the exchangeability at a fixed $\mathbf{u}$ that the calibration step requires holds only on the $a_i = 0$ fiber — a vanishing $2^{-k} = m^{-\Theta(1)}$ fraction of Valiant–Vazirani columns — and the “local comparator dominates the global decoder” calibration is not valid as stated. The structural facts, together with a finite two-point success-domination counterexample, are formalized with no `sorry` in the Lean module `Mettapedia.Computability.PNP.PostSwitchInputObstruction`. We further machine-check that the natural repair — re-deriving success domination on the T_i -invariant reduced projection $(\mathbf{z}, a_i)$ — does not reduce the problem: on that neutral interface, success domination is logically equivalent to the average-case witness-bit hardness the route must prove, so it renames the target rather than weakening it (stated conditional on two explicit, unproven hypotheses). We state the scope exactly: what is machine-checked is the *structural* input-preservation fact; the full claim that *no* local comparator dominates a global decoder on this ensemble engages the genuine difficulty and is *not* established by this work. We make no claim about P versus NP , which remains open; the formalization asserts no separation.

1 The problem and Ben’s quantale / Switching-by-Weakness route

1.1 The quantale-weakness route

The question is whether $P = NP$. Ben Goertzel’s manuscript $P \neq NP$: *A Non-Relativizing Proof via Quantale Weakness and Geometric Complexity*, version v12 (hereafter “v12”), proposes a distributional separation built from three interacting ingredients: a compositional *weakness* calculus

measuring the shortness of algorithms, the symmetry and sparsity of masked random 3-CNFs, and a genuine *algorithmic switching* statement turning any short polynomial-time decoder into a local per-bit rule on a constant fraction of blocks.

The weakness quantale. v12 measures shortness by the polytime-capped conditional description length

$$\text{weakness}(z \mid y) := K^{\text{poly}}(z \mid y),$$

used as the cost object of a quantale: costs add under composition and under independent block product (v12, Lemma A.8). The strategy is to oppose a *global budget* accrued by any short decoder across $t = \Theta(m)$ independent blocks against the constant-length per-block encoder available under a $P = NP$ self-reduction.

The masked-and-isolated ensemble. For a fixed clause density $\alpha > 0$ and $k = c_1 \log m$, a block is sampled (v12, §3.1) as $\Phi = (F^h, A, b)$, where: F is a constant-density random 3-CNF; a fresh mask $h = (\pi, \sigma) \in H_m = S_m \times (\mathbb{Z}_2)^m$ permutes variable names and flips literal signs to give F^h ; and a Valiant–Vazirani isolation layer adds a matrix $A \in \{0, 1\}^{k \times m}$ with pairwise-independent columns and a δ -biased right-hand side $b \in \{0, 1\}^k$. The block distribution $\mathcal{D}_m$ is the law of Φ conditioned on the event $\text{Unq}(\Phi)$ that Φ has a *unique* satisfying assignment $X = x(\Phi) \in \{0, 1\}^m$. Writing $a_i := Ae_i$ for the i -th column of A , the pair (a_i, b) is the *VV label* for bit i .

Local inputs and the post-switch interface. A sign-invariant SILS feature vector $\mathbf{z} := \text{feat}(F^h) \in \{0, 1\}^{r(m)}$ with $r(m) = O(\log m)$ is published, and v12 defines (Definition 3.10) the *per-bit local input*

$$\mathbf{u}_i(\Phi) := (\mathbf{z}(F^h), a_i = Ae_i, b) \in \{0, 1\}^{O(\log m)}, \quad (1)$$

together with the local σ -field $\mathcal{L}_i$ it generates.

Switching-by-Weakness. The structural core (v12, Theorem 4.2) is a normal form: for every polynomial-time decoder P of description length $\leq \delta t$ there is a short wrapper W with $|W| \leq |P| + O(\log m + \log t)$ and a subset $S \subseteq [t]$ with $|S| \geq \gamma t$ such that

$$(P \circ W)(\Phi)_{j,i} = h_{j,i}(\mathbf{u}_i(\Phi_j)), \quad j \in S, i \in [m], \quad (2)$$

for Boolean maps $h_{j,i} : \{0, 1\}^{O(\log m)} \rightarrow \{0, 1\}$, each computable in time $\text{poly}(\log m)$. The wrapper is constructed by ERM distillation (Proposition A.5), trained on *symmetrized, back-mapped* surrogate labels rather than on the truth X ; the link from the surrogate labels back to truth is the *calibration* Lemma 4.8.

Neutrality, sparsification, and the clash. Two orthogonal mechanisms then force near-randomness of every $\mathbf{u}$ -measurable per-bit rule on $\Theta(t)$ blocks: a sign-invariant *neutrality* lemma (Theorem 5.1), $\Pr[X_i = 1 \mid \mathcal{I}] = \frac{1}{2}$ for any sign-invariant view $\mathcal{I}$; and a *template sparsification* theorem (Theorem 5.10) showing that, at radius $r = c_3 \log m$, any fixed local per-bit rule is ε -high-bias on at most $o(t)$ blocks. Combined with single-block lower bounds for tiny ACC^0 /streaming decoders and independence across blocks (Theorem 6.7), one obtains a per-program small-success bound $2^{-\Omega(t)}$, hence the tuple incompressibility $K^{\text{poly}}((X_1, \dots, X_t) \mid (\Phi_1, \dots, \Phi_t)) \geq \eta t$ with high probability (Theorem 6.8). Under $P = NP$, a uniform constant-length program recovers each witness by bit-fixing with a USAT decider, so the same quantity is $\leq C$ (Proposition 7.2); for large t the two are incompatible, which v12 reads as $P \neq NP$ (Theorem 7.4).

1.2 The calibration crux

Every quantitative step downstream of (2) bounds the success of the *comparator* $P \circ W$ and transfers that bound back to P by *success domination* (v12, Corollary 4.6):

$$\Pr[P(\Phi) = X] \leq \Pr[(P \circ W)(\Phi) = X] + m^{-\Omega(1)}. \quad (3)$$

The comparator is trained on symmetrized surrogate labels Y_i , so its analysis is only meaningful once one knows that the Bayes-optimal predictor for the surrogate Y_i at a fixed local input $\mathbf{u}$ is *also* optimal for the truth X_i . This is the calibration Lemma 4.8, and its proof rests on a single exchangeability claim. In v12’s words (the “Calibration in one line” of §4.1, proved in Appendix A.5):

Fix $\mathbf{u} = (\mathbf{z}, a_i, b)$. The promise-preserving involution T_i bijects $(X_i = 0, Y_i = y) \leftrightarrow (X_i = 1, Y_i = 1 - y)$ without changing $\mathbf{u}$ or the measure. Thus $(X_i, Y_i) \mid \mathbf{u}$ is exchangeable, so $\Pr[X_i = 1 \mid \mathbf{u}] = \Pr[Y_i = 1 \mid \mathbf{u}] = f_i(\mathbf{u})$, and the Bayes rule $h_i^(\mathbf{u}) = \mathbf{1}[f_i(\mathbf{u}) \geq \frac{1}{2}]$ is optimal for both.*

The involution invoked here is the promise-preserving map of v12, Lemma 3.6,

$$T_i : (F^h, A, b) \mapsto (F^{\tau_i h}, A, b \oplus Ae_i), \quad (4)$$

where $\tau_i \in H_m$ flips only variable i ’s literal signs. It toggles the witness bit X_i and fixes any sign-invariant view $\mathcal{I}$, which is exactly the content of neutrality. The calibration argument additionally asserts that T_i acts on the *local input* $\mathbf{u}_i = (\mathbf{z}, a_i, b)$ “without changing $\mathbf{u}$.”

The detailed proof in Appendix A.5 (Lemma A.17) makes the action on each coordinate of $\mathbf{u}$ explicit. Its Step 1 lists the properties of the same map (4): it “preserves the SILS features $\mathbf{z}$ (which are sign-invariant),” “preserves a_i but *flips b by a_i* ,” and “preserves the uniqueness promise.” Step 2 then conditions on a fixed $\mathbf{u} = (\mathbf{z}, a_i, b)$ — a triple whose *third* coordinate is b — and concludes exchangeability of (X_i, Y_i) “at fixed $\mathbf{u}$.” These two statements are in tension: a map that flips b by a_i does not fix the triple $(\mathbf{z}, a_i, b)$ unless $a_i = 0$. It is precisely this bridge point that we formalize.

2 The formalized obstruction

We formalize the action of the involution (4) on the post-switch input (1), isolating exactly which part of $\mathbf{u}$ is invariant. The development is elementary and carries no `sorry`; it lives in the Lean module `Mettapedia.Computability.PNP.PostSwitchInputObstruction`, over a small bit-vector layer in which `BitVec n` is a Boolean function on `Fin n`, `zeroVec` is the all-false vector, and the Valiant–Vazirani right-hand-side update is the coordinatewise exclusive-or

```
def vvToggle {n : Nat} (a b : BitVec n) : BitVec n :=
  fun i => Bool.xor (b i) (a i)
```

which is the XOR $b \oplus a$ appearing in (4).

2.1 The post-switch input and the involution’s action

The post-switch input $\mathbf{u} = (\mathbf{z}, a_i, b)$ of (1) is the structure `PostSwitchInput`; the involution’s induced action `tiInputMap` is exactly (4) read on $\mathbf{u}$ (the SILS coordinate $\mathbf{z}$ held fixed by sign-invariance, the column a held fixed, and the VV right-hand side b toggled); and `invariantProjection` is the coordinate one obtains by *dropping* b .

```

structure PostSwitchInput (Z : Type*) (k : ℕ) where
  z : Z
  a : BitVec k
  b : BitVec k

/-- The exact local-input action induced by the paper's involution 'T_i'. -/
def tiInputMap {Z : Type*} {k : ℕ} (u : PostSwitchInput Z k) : PostSwitchInput Z k :=
  ⟨u.z, u.a, vvToggle u.a u.b⟩

/-- The obvious 'T_i'-invariant projection obtained by dropping the VV
right-hand side. -/
def invariantProjection {Z : Type*} {k : ℕ} (u : PostSwitchInput Z k) : Z × BitVec k :=
  (u.z, u.a)

```

The reduced projection (z, a_i) is always T_i -invariant. This is the part of $\mathbf{u}$ that the calibration argument may legitimately condition on.

```

@[simp] theorem invariantProjection_tiInputMap
  {Z : Type*} {k : ℕ} (u : PostSwitchInput Z k) :
  invariantProjection (tiInputMap u) = invariantProjection u := by
  rfl

```

The full input (z, a_i, b) , however, is fixed by T_i exactly on the zero-column fiber. The pivot lemma is that toggling b by a returns b iff a is the zero vector,

```

lemma vvToggle_eq_self_iff_zero {n : ℕ} (a b : BitVec n) :
  vvToggle a b = b ↔ a = zeroVec

```

from which the full-input preservation criterion follows immediately:

```

theorem tiInputMap_eq_self_iff_zeroColumn
  {Z : Type*} {k : ℕ} (u : PostSwitchInput Z k) :
  tiInputMap u = u ↔ u.a = zeroVec := by
  cases u
  simp [tiInputMap, vvToggle_eq_self_iff_zero]

```

2.2 The bridge point, stated as a single theorem

Combining the two facts gives the exact obstruction at the calibration bridge: on every nonzero VV column the reduced projection is preserved, yet the full input is not, because b moves.

```

/-- The exact calibration step cannot keep both the invariant projection and
the full post-switch input fixed on a nonzero VV column: '(z, a)' is preserved,
but 'b' changes and hence the full input changes. -/
theorem calibrationFullInputPreservation_fails_of_nonzeroColumn
  {Z : Type*} {k : ℕ} (u : PostSwitchInput Z k)
  (ha : nonzeroColumn u.a) :
  invariantProjection (tiInputMap u) = invariantProjection u ∧
  tiInputMap u ≠ u ∧ (tiInputMap u).b ≠ u.b := by
  exact ⟨invariantProjection_tiInputMap u,
    tiInputMap_ne_self_of_nonzeroColumn u ha,
    tiInputMap_changes_b_of_nonzeroColumn u ha⟩

```

As a global no-go form: in every positive VV width and for every concrete $\mathbf{z}$ -value, the proposition that T_i preserves the *full* local input on all nonzero columns is false.

```
theorem not_calibrationFullInputPreservation_on_nonzeroColumns
  {Z : Type*} (z : Z) {k : ℕ} (hk : 0 < k) :
  ¬ (∀ u : PostSwitchInput Z k, nonzeroColumn u.a -> tiInputMap u = u)
```

Why this voids the exchangeability at fixed $\mathbf{u}$. The calibration of §1.2 derives the symmetry of the conditional law of (X_i, Y_i) at a fixed $\mathbf{u} = (\mathbf{z}, a_i, b)$ from the bijection T_i . But by (4) and `tiInputMap`, T_i sends the local input $\mathbf{u} = (\mathbf{z}, a_i, b)$ to $\mathbf{u}' = (\mathbf{z}, a_i, b \oplus a_i)$. When $a_i \neq 0$ these are *different* points of the local alphabet: the involution exchanges configurations across the two fibers $\{b\}$ and $\{b \oplus a_i\}$, not within the single fiber $\{b\}$ on which the calibration conditions. The symmetry it supplies is therefore a symmetry of the law conditioned on the reduced projection $(\mathbf{z}, a_i)$ — equivalently, averaged over the b -coordinate — and not of the law at fixed full $\mathbf{u}$. The exchangeability “ $(X_i, Y_i) \mid \mathbf{u}$ is exchangeable” holds, via this involution, only on the locus $a_i = 0$.

The $a_i = 0$ fiber is vanishing. The column $a_i = Ae_i$ is the i -th column of the Valiant–Vazirani matrix A , whose columns are pairwise independent and marginally uniform in $\{0, 1\}^k$ (v12, Definition 3.3). Hence $\Pr[a_i = 0] = 2^{-k}$, and with $k = c_1 \log m$ this is $m^{-\Theta(1)}$ — a polynomially vanishing fraction of columns. The promise-preserving exchangeability that the calibration step needs at fixed $\mathbf{u}$ is thus available only on a vanishing fraction of the VV columns that index the per-bit rules, exactly the regime the lower bound must control.

2.3 A finite success-domination counterexample

The same module records, in the smallest nontrivial setting, why “two instances that share a local input” cannot be calibrated by a $\mathbf{u}$ -measurable rule: a global decoder that reads the full instance can strictly out-predict every rule that depends only on the shared local input. Take two equally weighted full instances (encoded by a Boolean) that map to the *same* nonzero one-bit post-switch input. A global decoder recovers the witness on both; any local decoder, forced to use only the common input, is a fixed guess and is correct on at most one.

```
theorem twoPointGlobalSuccessCount_eq_two :
  twoPointGlobalSuccessCount = 2 := by rfl

theorem twoPointLocalSuccessCount_le_one (guess : Bool) :
  twoPointLocalSuccessCount guess ≤ 1 := by
  cases guess <|> decide

/-- Finite domination counterexample: the global decoder succeeds on both
full instances, while every decoder depending only on the shared local input
succeeds on at most one. -/
theorem finite_successDomination_counterexample :
  twoPointGlobalSuccessCount = 2 ∧
  (∀ guess : Bool, twoPointLocalSuccessCount guess ≤ 1) ∧
  (∀ guess : Bool, twoPointLocalSuccessCount guess < twoPointGlobalSuccessCount)
```

This is a finite witness that success domination (3) cannot be obtained *from sharing $\mathbf{u}$ alone*: when two outcomes collapse to the same local input but differ in the truth, the comparator that

is forced through $\mathbf{u}$ loses to the global decoder. It is the local, combinatorial shadow of the fiber-crossing exhibited above; the involution that would have justified calibration moves $\mathbf{u}$ off the fiber rather than within it.

2.4 The reduced-projection repair is circular

The natural repair conditions the local rules on the T_i -invariant projection $(\mathbf{z}, a_i)$ — the coordinate calibration may legitimately use — rather than on the full $\mathbf{u}$, hoping to trade the calibration gap for a coarser interface. We machine-check that this does not reduce the problem.

Write StarSW for the half-bound success domination must deliver on a neutral interface: that every decoder scores at most $\frac{1}{2} + \text{slack}$ on the witness bit — exactly the average-case witness-bit hardness of $\mathcal{D}_m$. If the reduced projection is neutral — the property the repair itself asserts — then the best $(\mathbf{z}, a_i)$ -measurable comparator scores exactly $\frac{1}{2}$, and success domination on $(\mathbf{z}, a_i)$ is *logically equivalent* to StarSW:

```
theorem bestLocalSuccess_eq_half_of_neutral (hN : Neutral u x) :
  bestLocalSuccess u x = 1 / 2

theorem coarseDominationAll_iff_starSW_of_neutral (hN : Neutral u x) :
  CoarseDominationAll u x decoderSuccess slack
  <-> StarSW decoderSuccess slack
```

Re-deriving domination on $(\mathbf{z}, a_i)$ is thus re-deriving the hardness target, not reducing to anything weaker. A concrete two-point instance makes the failure explicit: two full instances that share the reduced projection but differ in b give a neutral projection on which a b -reading decoder wins on both, so domination fails ($1 \leq \frac{1}{2}$):

```
theorem coarseProjectionToy_not_dominates :
  not (Dominates coarseProjectionToyUCoarse coarseProjectionToyWitness ...)
```

This is an instance of a *no-threading* impossibility: a neutral statistic cannot also be sufficient for a target that is a deterministic function of the instance (`neutral_not_sufficient`, from the general `noThreadingLemma`). Every coordinate dropped to gain T_i -invariance is a coordinate the global decoder may still use; neutrality and sufficiency trade one for one.

We are exact about scope. The conclusion that the repair is circular *on the ensemble* is a theorem *conditional on two explicit hypotheses we do not prove* — that $(\mathbf{z}, a_i)$ is neutral on $\mathcal{D}_m$, and that StarSW implies the separation — both kept in the signature:

```
theorem coarseProjectionRepair_obstruction_conditional
  (EnsembleCoarseNeutral : Neutral u x)
  (StarSWImpliesSeparation : StarSW decoderSuccess slack -> pNeNPClaim) :
  (CoarseDominationAll u x decoderSuccess slack <-> StarSW decoderSuccess slack)
  /\ (CoarseDominationAll u x decoderSuccess slack -> pNeNPClaim)
```

The machine-checked content is the *equivalence* and the *finite counterexample*; together they show the reduced-projection repair renames the average-case hardness rather than reducing to it. The module is `Mettapedia.Computability.PNP.CoarseProjectionObstruction`, without sorry.

3 Status and scope

We state the scope of this finding exactly.

What is established. The action of the promise-preserving involution T_i of (4) on the post-switch input $\mathbf{u}_i = (\mathbf{z}, a_i, b)$ of (1) is $b \mapsto b \oplus a_i$; it preserves the reduced projection $(\mathbf{z}, a_i)$ (theorem `invariantProjection_tiInputMap`) and preserves the full input $(\mathbf{z}, a_i, b)$ *if and only if* $a_i = 0$ (theorems `tiInputMap_eq_self_iff_zeroColumn` and `calibrationFullInputPreservation_fails_of_nonzeroColumn`); on every nonzero column the third coordinate b genuinely moves, so the full-input preservation invoked by the calibration argument fails (`not_calibrationFullInputPreservation_on_nonzeroColumns`). A finite two-point success-domination counterexample (`finite_successDomination_counterexample`) shows that a $\mathbf{u}$ -measurable comparator cannot dominate a global decoder merely because they share the local input. All of this is machine-checked, without `sorry`, in `Mettapedia.Computability.PNP.PostSwitchInputObstruction`. The consequence is precise: the exchangeability of (X_i, Y_i) at a fixed $\mathbf{u}$ that the calibration step (v12, Lemma 4.8 / Appendix A.5, Lemma A.17) derives from T_i holds, via that involution, only on the $a_i = 0$ fiber, a vanishing $2^{-k} = m^{-\Theta(1)}$ fraction of VV columns. The local-comparator-dominates- global-decoder calibration is therefore not valid as stated.

What is not established. This is an obstruction to the calibration *as written*, not a refutation of its conclusion. We have machine-checked the *structural* fact that the cited involution does not fix the full local input off the zero-column fiber; we have *not* established that no $\mathbf{u}$ -measurable comparator can dominate a global decoder on the ensemble $\mathcal{D}_m$. That stronger, measure-theoretic statement — a genuine anti-concentration / independence claim about the conditional law of (X_i, Y_i) given the full $\mathbf{u}$, integrated against the Valiant–Vazirani and masking laws — is exactly the difficulty the proof must confront, and it is left open here. In particular, conditioning only on the invariant projection $(\mathbf{z}, a_i)$ does not by itself help: §2.4 shows the domination bound one would re-derive there is equivalent to the average-case hardness it must prove, and a *separate* neutral local mechanism for the b -coordinate is excluded by the same no-threading impossibility. What this work does *not* rule out is a domination bound from genuinely new structure — an exchangeability at fixed $\mathbf{u}$ from a symmetry other than T_i , or a direct anti-concentration of $(X_i, Y_i) \mid \mathbf{u}$ against the masking and Valiant–Vazirani laws.

P versus NP remains open. We make no claim, in either direction, about P versus NP. The finding concerns the validity of one step in one proposed route, not the truth of the separation. Consistent with this, the formalization itself asserts no separation: the project’s route-status record carries the verdict `claimsFinalSeparation = false`, and the module above proves a property of the calibration bridge, not a circuit or complexity-class lower bound.

4 What remains

The obstruction localizes the open question very precisely. The route’s lower bound is sound *given* the calibration link (3); the calibration link is justified, in v12, by an exchangeability that the cited involution supplies only on a vanishing fraction of columns. Closing the route therefore requires one of the following, none of which is provided here.

- *Calibration on the reduced projection — machine-checked to be circular.* One might re-derive the success-domination bound conditioning the local rules on the T_i -invariant projection $(\mathbf{z}, a_i)$ rather than the full $\mathbf{u} = (\mathbf{z}, a_i, b)$. Section 2.4 shows this is not a genuine reduction: on the neutral reduced projection, success domination is *equivalent* to the average-case hardness StarSW the route must prove (conditional on the two explicit hypotheses of §2.4), so it renames the target rather than weakening it. We record it here only to mark this door closed.
- *Exchangeability at fixed $\mathbf{u}$ from new structure.* Exhibit a promise-preserving symmetry that genuinely fixes the full $(\mathbf{z}, a_i, b)$ while toggling X_i , or otherwise establish the anti-concentration of $(X_i, Y_i) \mid \mathbf{u}$ directly against the masking and Valiant–Vazirani laws — the measure-theoretic claim left open in §3.
- *A quantitative b -marginal control.* Bound the contribution of the b -coordinate to a global decoder’s advantage, so that the fiber-crossing of T_i costs only $m^{-\Omega(1)}$ and can be absorbed into the slack of (3).

The two latter items are substantive analytic tasks on the masked-and-isolated ensemble; the first is closed above, machine-checked to be circular rather than a reduction. Until one of the substantive tasks is carried out, the Switching-by-Weakness route does not, as written, establish its distributional lower bound, because the calibration step on which that bound depends is not valid at fixed $\mathbf{u}$ on the bulk of the ensemble. The machine-checked content of this paper is exactly the boundary of that gap, and nothing beyond it.

Provenance

The manuscript discussed is Ben Goertzel, $P \neq NP$: *A Non-Relativizing Proof via Quantale Weakness and Geometric Complexity*, version v12. The formalization is work of the MeTTapedia formalization effort; the Lean development is the module `Mettapedia.Computability.PNP.PostSwitchInputObstruction` (the calibration obstruction), together with `CoarseProjectionObstruction` and the general `NoThreadingLemma` (the reduced-projection circularity of §2.4); all carry no `sorry`. Notation in Sections 1–1.2 follows v12 (the ensemble $\mathcal{D}_m$, the masks H_m , the Valiant–Vazirani labels (a_i, b) with $a_i = Ae_i$, the SILS features $\mathbf{z}$, the post-switch input $\mathbf{u}_i = (\mathbf{z}, a_i, b)$, the involution T_i , and the polytime-capped weakness K^{poly}).